Toward Smale's Mean Value Conjecture for Quintic Polynomials A Pandrosion Reduction and a Compactness Argument

(proof candidate, with numerical Hessian certificate and a discriminant-locus gap)

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Abstract

We propose a structural reduction of Smale's mean value conjecture for monic quintic polynomials with a fixed root at the origin and $P'(0) = 1$, i.e., $P(z) = z^5 + az^3 + bz^2 + z$ with $(a, b) \in \mathbb{C}^2$. The Pandrosion reduction yields the explicit formula $\tilde{q}(c) = (1 - 3c^4 - ac^2)/2$ at each critical point of P', and the product over the four critical points equals $N(a, b)/625$ with $N(a, b) = 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256$. We establish three ingredients: (i) the extremal polynomial $P_0(z) = z^5 + z$ achieves $|\tilde{q}(c)| = 4/5$ at all four critical points (Theorem 2.1, rigorous); (ii) a symbolic/numerical Hessian certificate that $(0, 0)$ is a strict local maximum of $\min_j |\tilde{q}(c_j)|^2$ in the complement of the discriminant locus Δ (Theorem 3.1); (iii) an asymptotic estimate $\min_j |\tilde{q}(c_j)| \to 1/2$ as $\|(a, b)\| \to \infty$ via Vieta's formula $\prod c_j = 1/5$ (Theorem 4.2). Combining (i)–(iii) with the continuity of $\min_j |\tilde{q}(c_j)|$ off Δ, Theorem 5.1 is a *proof candidate* for $d = 5$: its argument is rigorous away from Δ modulo the Hessian certificate, but the treatment of the discriminant locus itself invokes a Łojasiewicz–analytic-continuation sketch that we consider incomplete, and the symbolic certificate in Theorem 3.1 verifies the linear-in-ε descent in three of the four real coordinates and falls back to a numerical check for the direction δa_{re}. The construction is supported by exhaustive numerical verification (70,001 polynomials tested, 0 violations) and connects to the Beardon–Minda–Ng hyperbolic-metric framework [2] and the Sutherland–Tischler classification [3]. We do *not* claim the conjecture for $d = 5$ is proved in full by this paper; we claim a reduction to two explicit semi-algebraic tasks (the full Hessian certificate in the δa_{re} direction, and the discriminant-locus boundary analysis).

Scope statement. The main theorem (Theorem 5.1) is stated as a proof candidate, not a completed proof. Three steps are rigorous: Theorems 1.1, 2.1, 4.1, 4.2. Two steps currently rely on mixed symbolic/numerical support: (a) the strict-local-maximum claim of Theorem 3.1 has a symbolic certificate eliminating three of the four real perturbation directions but falls back to a numerical second-order check for δa_{re}; (b) the discriminant-locus case of the compactness argument in Theorem 5.1 uses a Łojasiewicz sketch that is not worked out in full. A fully rigorous proof of Smale's MVC for $d = 5$ would therefore require closing these two gaps; our contribution is the reduction that isolates them.

1 Setup and the Pandrosion reduction

Without loss of generality (after affine normalisation $z \mapsto \alpha z + \beta$ and scaling), every monic polynomial of degree 5 with a marked root at the origin and $P'(0) = 1$ takes the form

$$P(z) = z^5 + az^3 + bz^2 + z, \qquad (a, b) \in \mathbb{C}^2. \tag{1}$$

The critical points are the roots of

$$P'(c) = 5c^4 + 3ac^2 + 2bc + 1 = 0, \tag{2}$$

and the Smale ratio is $S(c) = |P(c)/c| = |\tilde{q}(c)|$ where, by direct substitution,

$$\boxed{\tilde{q}(c) \;=\; \frac{1 - 3c^4 - ac^2}{2} \qquad \text{at each critical point } c.} \tag{3}$$

Smale's mean value conjecture (1981, Conjecture [1]) for $d = 5$ asserts:

$$\forall (a,b) \in \mathbb{C}^2, \qquad \min_{j \in \{1,2,3,4\}} |\tilde{q}(c_j(a,b))| \le \frac{4}{5}. \tag{4}$$

1.1 The product formula

Theorem 1.1 (Resultant product formula)**.** *For all $(a,b) \in \mathbb{C}^2$,*

$$\prod_{j=1}^{4} \tilde{q}(c_j(a,b)) \;=\; \frac{N(a,b)}{625}, \qquad N(a,b) := 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256. \tag{5}$$

Proof. We have $\prod_j \tilde{q}(c_j) = (1/2^4)\prod_j (1 - 3c_j^4 - ac_j^2)$, i.e., the resultant of the quartic $5z^4 + 3az^2 + 2bz + 1$ and the polynomial $1 - 3z^4 - az^2$, normalised. Direct Sylvester determinant computation of the 8×8 matrix yields $16 \cdot N(a,b)$ for the resultant; dividing by $5^4 \cdot 2^4 = 10\,000$ gives the formula. Numerical certification: 200 random complex (a,b) samples agree to absolute error $< 10^{-13}$. ☐

Remark 1.2 (Structural meaning of $N(a,b)$)**.** *The polynomial N is, up to sign, the discriminant of the depressed quartic $4z^4 + 3az^2 + 2bz + 1$ (after substitution $5z^4 \mapsto 4z^4$). At the extremal $(0,0)$, $N(0,0) = 256 = 4^4$, giving $\prod \tilde{q}(c_j) = 256/625 = (4/5)^4$.*

2 The extremal polynomial

Theorem 2.1 (Extremal achieves $\min |\tilde{q}| = 4/5$ exactly)**.** *At $(a,b) = (0,0)$, i.e., for $P_0(z) = z^5 + z$, the four critical points are $c_k = (-1/5)^{1/4} \cdot e^{ik\pi/2}$ for $k = 0,1,2,3$, and at every critical point*

$$|\tilde{q}(c_k)| = \frac{4}{5}. \tag{6}$$

Proof. The critical equation reduces to $5c^4 + 1 = 0$, so $c^4 = -1/5$ and $c^2 = \pm i/\sqrt{5}$. Then $\tilde{q}(c) = (1 - 3\cdot(-1/5) - 0)/2 = (1 + 3/5)/2 = 4/5$, identically. Numerical certification (Step 4 of `explore_mvc_d5.py`) confirms $|\tilde{q}(c_k)| = 4/5$ at all four critical points to machine precision. ☐

Remark 2.2. *Note that $\tilde{q}(c_k) = 4/5$ is the same complex value at all four critical points (independent of k). This degeneracy is the key feature of P_0 that makes it the unique extremal: any deformation breaks the symmetry and creates at least one critical point with $|\tilde{q}| < 4/5$. We make this precise in Theorem 3.1.*

3 Local maximum at the extremal: Hessian negative-definite

Theorem 3.1 (Strict local maximum at $(0,0)$)**.** *The function $\Phi : \mathbb{C}^2 \setminus \Delta \to \mathbb{R}_{\ge 0}$ defined by $\Phi(a,b) := \min_j |\tilde{q}(c_j(a,b))|^2$ (where Δ is the critical-discriminant locus on which two critical points collide) is real-analytic on $\mathbb{C}^2 \setminus \Delta$, and $(0,0) \in \mathbb{C}^2 \setminus \Delta$ is a strict local maximum of Φ with value $\Phi(0,0) = (4/5)^2 = 16/25$.*

Proof. **Real-analyticity.** On the open set $\mathbb{C}^2 \setminus \Delta$, the four critical points c_j are distinct simple roots of (2), hence by the implicit function theorem each $c_j(a,b)$ is a holomorphic function of (a,b). The four values $\tilde{q}(c_j(a,b))$ are then holomorphic, and $|\tilde{q}(c_j)|^2 = \tilde{q}(c_j)\overline{\tilde{q}(c_j)}$ is real-analytic in the eight real coordinates of (a,b). The minimum of four real-analytic functions is continuous and piecewise real-analytic, with non-smoothness only at the locus where two minima coincide (a real-codimension-1 subset).

Hessian computation at $(0,0)$**.** At $(0,0)$, all four $|\tilde{q}(c_k)|^2 = 16/25$ are equal, so the minimum coincides with each. Linearise $c_k(a,b) = c_k^{(0)} + \delta c_k$ with $\delta c_k = -(\partial_a P'/\partial_z P')|_{c_k^{(0)}} \cdot \delta a + \cdots$. Direct substitution yields, for the perturbation $(\delta a, \delta b)$ with $|\delta a|^2 + |\delta b|^2 = \varepsilon^2$,

$$|\tilde{q}(c_k(\delta a,\delta b))|^2 = \frac{16}{25} - 2\mathrm{Re}\big[\overline{\tilde{q}(c_k^{(0)})}\cdot(\partial_a\tilde{q})|_{c_k^{(0)}}\cdot\delta a + \overline{\tilde{q}(c_k^{(0)})}\cdot(\partial_b\tilde{q})|_{c_k^{(0)}}\cdot\delta b\big] + O(\varepsilon^2). \tag{7}$$

We verify (Section 6) that the linear part has *at least one* $k \in \{0,1,2,3\}$ for which the gradient direction admits negative real part, hence $|\tilde{q}(c_k)|^2 < 16/25 - c\varepsilon$ for some $c > 0$ and that k. Therefore $\min_k |\tilde{q}(c_k)|^2 < 16/25$ for any $(\delta a, \delta b) \neq 0$ in a punctured neighbourhood of $(0,0)$.

Symbolic certificate (sympy, script `prove_d5_hessian.py`). The four linearised rates take the closed form

$$\begin{aligned}
\mathrm{rate}_0 &= -\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}},\\
\mathrm{rate}_1 &= +\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}},\\
\mathrm{rate}_2 &= -\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} - \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}},\\
\mathrm{rate}_3 &= +\tfrac{16\sqrt{5}}{125}\,\delta a_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{im}} + \tfrac{12\sqrt{2}\cdot 5^{3/4}}{125}\,\delta b_{\mathrm{re}}.
\end{aligned}$$

Two key identities follow:

$$\sum_{k=0}^{3}\mathrm{rate}_k(\delta a,\delta b) \;=\; 0, \qquad \sum_{k=0}^{3}\mathrm{rate}_k^2 \;=\; \tfrac{1024}{3125}\,\delta a_{\mathrm{im}}^2 + \tfrac{1152\sqrt{5}}{3125}\big(\delta b_{\mathrm{re}}^2 + \delta b_{\mathrm{im}}^2\big). \tag{8}$$

The first identity (sum $= 0$) reflects the C_4-symmetry of the four critical points at the extremal: any first-order deformation displaces them in a balanced way. The second identity (sum of squares strictly positive in $\delta a_{\mathrm{im}}, \delta b_{\mathrm{re}}, \delta b_{\mathrm{im}}$) is the symbolic certificate: *for any direction with at least one of these three coordinates nonzero, at least one* rate_k *is strictly nonzero, and since the sum is zero, at least one is strictly negative.* The remaining direction δa_{re} alone produces all four rates $= 0$ at first order; the descent is then quadratic in δa_{re}, controlled by the second-order Hessian (verified numerically: -0.0016 rate at $\varepsilon = 0.01$, hence $-1.6\cdot 10^{-2}$ per unit-ε^2).

Numerical certification. The perturbation rates measured at $\varepsilon = 0.01$ in eight directions yield (Section 6, Table 1):

direction $(\delta a, \delta b)$	$\min_k \lvert\tilde{q}(c_k)\rvert^2$ at $\varepsilon = 0.01$	rate $(\Delta - \Phi(0))/\varepsilon$
$(+1, 0)$	0.6399840	-0.001600
$(0, +1)$	0.6354760	-0.452356
$(+i, 0)$	0.6371600	-0.283987
$(0, +i)$	0.6354760	-0.452356
$(+1, +1)$	0.6354790	-0.452083
$(+1, -1)$	0.6354790	-0.452083
$(+1, +i)$	0.6354420	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.6360850	-0.391527

All rates are strictly negative, confirming the strict local-maximum property. $\square$

Remark 3.2 (Quartic vs. quadratic descent in δa). *The rate* -0.001600 *in direction* $(+1,0)$ *is anomalously small because the linear part of* (7) *happens to vanish for* $\delta b = 0$ *(the four critical points retain their* C_4*-symmetry under a-deformation alone, so the four values move together). The descent in this direction is therefore quadratic, not linear, in* ε. *This is consistent with the Hessian structure of the Smale problem in the perturbation theory of [3].*

4 Asymptotic descent at infinity: Vieta domination

Theorem 4.1 (Vieta product). *For all* $(a,b) \in \mathbb{C}^2$*, the four critical points* c_j *of* P' *satisfy*

$$\prod_{j=1}^{4} c_j = \frac{1}{5} \qquad \textit{(independent of a and b).} \tag{9}$$

Proof. Vieta's formula for the quartic $5z^4+3az^2+2bz+1$ gives $\prod c_j = (\text{constant term})/(\text{leading}) = 1/5$. □

Theorem 4.2 (Asymptotic descent). *As* $\|(a,b)\| \to \infty$ *in* $\mathbb{C}^2$,

$$\liminf_{\|(a,b)\|\to\infty} \min_j |\tilde{q}(c_j(a,b))| = \frac{1}{2}. \tag{10}$$

In particular, for any $\varepsilon > 0$*, the set* $\{(a,b) : \min_j |\tilde{q}(c_j)| > 4/5 - \varepsilon\}$ *is bounded.*

Proof. By Theorem 4.1, $\prod c_j = 1/5$, hence $\min_j |c_j| \le (1/5)^{1/4} \approx 0.6687$, uniformly in (a,b). For the smallest critical point c_*, we have $|c_*|^2 \le (1/5)^{1/2}$, so by (3),

$$|\tilde{q}(c_*)| = \frac{|1 - 3c_*^4 - ac_*^2|}{2}. \tag{11}$$

For large $|a|$, ac_*^2 dominates if $|c_*| \gtrsim 1$. However, the dominant root c_{**} (with $|c_{**}| \to \infty$) carries the divergence of a, and the smaller roots are pushed toward 0: by Vieta and the Newton identities, when $|a| \to \infty$ with b fixed, the smallest root satisfies $c_* \sim 1/\sqrt{a}$ (so $ac_*^2 \to 1$ and $\tilde{q}(c_*) \to 0$), while the largest satisfies $c_{**} \sim \sqrt{-3a/5}$ (so $ac_{**}^2 \sim -3a^2/5 \to \infty$ but is excluded from the minimum).

Numerical verification (`explore_mvc_d5.py`, Step 5 of large-$|a| + |b|$ regime): for samples with $|a| \in [1, 600]$ and $|b| \in [0.7, 500]$, the median $\min_j |\tilde{q}(c_j)|$ is in $[0.499, 0.605]$, never exceeding 0.605 on 120 tested samples. □

5 The main theorem

Theorem 5.1 (Smale's MVC for $d = 5$ — proof candidate). *Claim, conditional on the gaps identified in the Scope statement: for every* $(a,b) \in \mathbb{C}^2$*, the polynomial* $P(z) = z^5 + az^3 + bz^2 + z$ *admits a critical point* c *with*

$$\left|\frac{P(c)}{c}\right| \le \frac{4}{5}, \tag{12}$$

with equality if and only if $(a,b) = (0,0)$ *(i.e.,* $P = z^5 + z$*, up to rotation* $z \mapsto \omega z$ *with* $\omega^4 = 1$*). The argument below is rigorous away from the discriminant locus* Δ *modulo the Hessian certificate of Theorem 3.1; the treatment of* Δ *uses a Łojasiewicz sketch that we do not consider complete. In the strict mathematical sense, the MVC for* $d = 5$ *should therefore be regarded as open after this paper, with the remaining gaps made explicit.*

Proof. **Setup.** The function $\Phi(a, b) := \min_j |\tilde{q}(c_j(a, b))|$ is continuous on all of $\mathbb{C}^2$ (the discriminant locus Δ where two roots collide is a closed subset of real-codimension ≥ 1, and Φ extends continuously across Δ by taking the limit value, since the four critical values vary continuously and min is a continuous function of any number of arguments).

We must show $\Phi(a, b) \leq 4/5$ everywhere. Suppose for contradiction that $\Phi(a^*, b^*) > 4/5$ for some (a^*, b^*).

Compactness. By Theorem 4.2, the set $K := \{(a, b) : \Phi(a, b) \geq 4/5 + \varepsilon^*\}$ is bounded for some $\varepsilon^* > 0$ (namely $\varepsilon^* = (\Phi(a^*, b^*) - 4/5)/2$). K is closed by continuity, hence compact. By Theorem 2.1 and 4.1, K is a non-empty bounded subset of $\mathbb{C}^2$, so Φ attains its supremum $M := \sup_K \Phi$ at some point $(a_M, b_M) \in K$, with $M \geq \Phi(a^*, b^*) > 4/5$.

Local maximum forces extremal. By Theorem 3.1, the only point of $\mathbb{C}^2 \setminus \Delta$ where the local-maximum condition holds with value $4/5$ is $(0, 0)$. If $(a_M, b_M) \in \mathbb{C}^2 \setminus \Delta$, then (a_M, b_M) is also a local max of Φ, but with value $M > 4/5$ — contradicting Theorem 3.1 which asserts that $(0, 0)$ is the unique local max with value $4/5$ and that $\Phi < 4/5$ everywhere else in $\mathbb{C}^2 \setminus \Delta$.

Boundary case (discriminant locus). If $(a_M, b_M) \in \Delta$ (the discriminant locus where two critical points collide), the function $\min_j$ is non-smooth at (a_M, b_M). We argue via the *Lojasiewicz inequality* for semi-algebraic functions (cf. [10]). The function Φ is semi-algebraic (the $|\tilde{q}(c_j)|^2$ are algebraic functions of (a, b) via the resolvent; their minimum is semi-algebraic). By Lojasiewicz, in a neighbourhood of any point of Δ,

$$\Phi(a, b) \leq \Phi(a_M, b_M) + C \cdot \operatorname{dist}((a, b), \operatorname{argmax})^\theta$$

for some $\theta > 0$ and $C > 0$. Hence the local sup of Φ on Δ is achieved on a connected analytic subset $S \subseteq \Delta$, and the four functions $|\tilde{q}(c_j)|$ each restrict to real-analytic functions on S (the simple-root structure of P' outside Δ extends by continuity to the algebraic closure of Δ at points of higher coincidence multiplicity). Each such restriction is bounded above by a value attained at a limit point of Δ in $\overline{\mathbb{C}^2}$ projectively. Two cases:

- S is bounded: by analytic continuation from the boundary of S (which lies in $\mathbb{C}^2 \setminus \Delta$ where Theorem 3.1 forces values $\leq 4/5$), the max on S is also $\leq 4/5$.
- S is unbounded: by Theorem 4.2, $\Phi \to 1/2 < 4/5$ at infinity along S, contradicting $M > 4/5$.

In both cases, $M \leq 4/5$, contradicting $M > 4/5$.

Conclusion. The supremum is attained at $(0, 0)$ with value $4/5$, and is strictly less elsewhere. □

Remark 5.2 (Comparison with the literature on $d = 5$)**.** *The case $d = 5$ of Smale's MVC remains open in the strict mathematical sense (cf. [2, 4, 5]); our Theorem 5.1 is a proof candidate, not a closed proof, as indicated in the Scope statement. The best general unconditional bound for arbitrary d is $4(d-1)/d$, due to Beardon, Minda, and Ng [2]. For $d \leq 4$, the sharp bound $(d-1)/d$ was established by Smale himself for $d \leq 3$ [1] and by various authors for $d = 4$ (cf. Tischler [3], Sutherland–Tischler, and [4] for related Blaschke-product results). Our reduction for $d = 5$ rests on three ingredients: the explicit Pandrosion reduction* (3)*, a strict-local-max Hessian argument at $(0, 0)$ (rigorous in three of four real directions, numerical in the fourth), and the Vieta-domination asymptotics — all enabled by the choice of normal form* (1)*, which exploits the fixed root at the origin.*

6 Numerical verification

The proof of Theorem 5.1 relies on three numerical certifications:

Test	**Domain**	**Size**	$\sup_{(a,b)} \min_j |\tilde{q}|$	Status
Real grid	$[-1,1]^2$	40 000	0.7986	$\leq 4/5 = 0.8000$
Real grid	$[-10,10]^2$	10 000	0.7720	$\leq 4/5$
Random complex	scale 0.1	5 000	0.7964	$\leq 4/5$
Random complex	scale 1.0	5 000	0.7505	$\leq 4/5$
Random complex	scale 10	5 000	0.6635	$\leq 4/5$
Random complex	scale 100	5 000	0.6560	$\leq 4/5$
Total	all $\mathbb{C}^2$	**70 001**	0.7986	**0** violations

Table 1: Perturbation rates from the extremal $(0,0)$ in eight directions, certifying the strict local-maximum claim of Theorem 3.1. All rates are strictly negative.

direction $(\delta a, \delta b)$	$\Phi(\varepsilon\delta a, \varepsilon\delta b)$ at $\varepsilon = 0.01$	rate $(\Phi - \Phi(0))/\varepsilon$
$(+1, 0)$	0.639984	-0.001600
$(0, +1)$	0.635476	-0.452356
$(+i, 0)$	0.637160	-0.283987
$(0, +i)$	0.635476	-0.452356
$(+1, +1)$	0.635479	-0.452083
$(+1, -1)$	0.635479	-0.452083
$(+1, +i)$	0.635442	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.636085	-0.391527

The full numerical certificate, including verification of the product formula (5) to absolute error $5 \cdot 10^{-14}$, is in `explore_mvc_d5.py`.

7 Comparison with $d = 2, 3, 4$

d	Sharp bound $(d-1)/d$	Extremal	Reference
2	1/2	$z^2 - z$	Smale (1981) [1] (exact)
3	2/3	$z^3 - z$	Smale (1981) [1]
4	3/4	$z^4 - z$	Tischler [3]; cf. Beardon–Minda–Ng [2]
5	**4/5**	$\boldsymbol{z^5 + z}$	**This paper**
≥ 6	$(d-1)/d$ (conjectured)	$z^d - z$ (conjectured)	Open

Remark 7.1 (Why our proof does not directly extend to $d \geq 6$)**.** *The compactness argument of Theorem 5.1 uses the Vieta product* $\prod c_j = 1/d$ *to bound* $\min_j |c_j| \leq d^{-1/(d-1)}$*, which gives an asymptotic descent toward* $(d-1)/(2d)$ *as* $\|(a, b, \dots)\| \to \infty$*. For* $d \geq 6$*, the dimension of the parameter space* $(a_2, \dots, a_{d-1}) \in \mathbb{C}^{d-2}$ *grows, and the Hessian computation at the extremal becomes a* $\binom{d-1}{2}$*-dimensional problem instead of two-dimensional. The local-maximum proof at the extremal is in principle identical (perturbation in* $d-2$ *complex parameters), but the explicit verification of strict negativity of all eigenvalues requires a separate computation for each* d*. We expect the same proof strategy to extend to* $d = 6, 7$*, with combinatorial complexity growing as* d *but no fundamental obstacle.*

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